Software update manual – V1.5.x

Additional steps required to migrate from V1.4.x to V1.5.x

1. Introduction

With the introduction of ePU10 software version 1.5.x, non-compatible changes have been added to the workflow. This creates additional requirements to the update workflow of any system operating on an older release version. The standard update sequence will result in a non-operational machine.

The incompatibility of this update is only on software and settings level. All machines operating 1.4.x software are capable of running 1.5.0 after updating the required subsystems.

Compared to the regular software update procedure, the 1.5.0 update has these additional steps:

- Updating Vacon inverter settings, to be performed before updating the PLC.
- Correcting Twincat PLC card configuration, to be performed before updating the PLC.
- Check ePU10 hardware configuration settings, to be performed after updating the PLC

All update steps assume:

- System out of operation with external supply and/or sufficient UPS SOC (>70%)
- Update laptop physically wired to ePU10 internal network or connected via an RMS VPN tunnel.

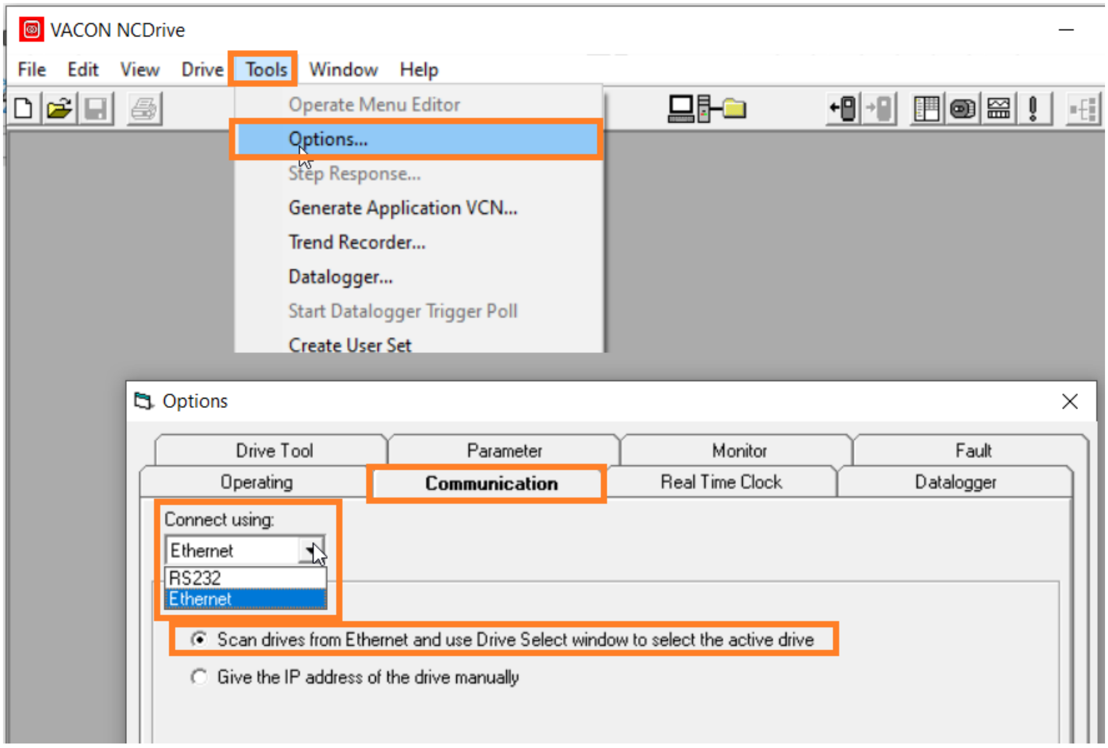
2. Vacon settings update

Before updating the PLC software, the inverter settings need to be updated. If the update is performed remotely, first install NCdrive on the PLC. Otherwise skip to step 2.2.

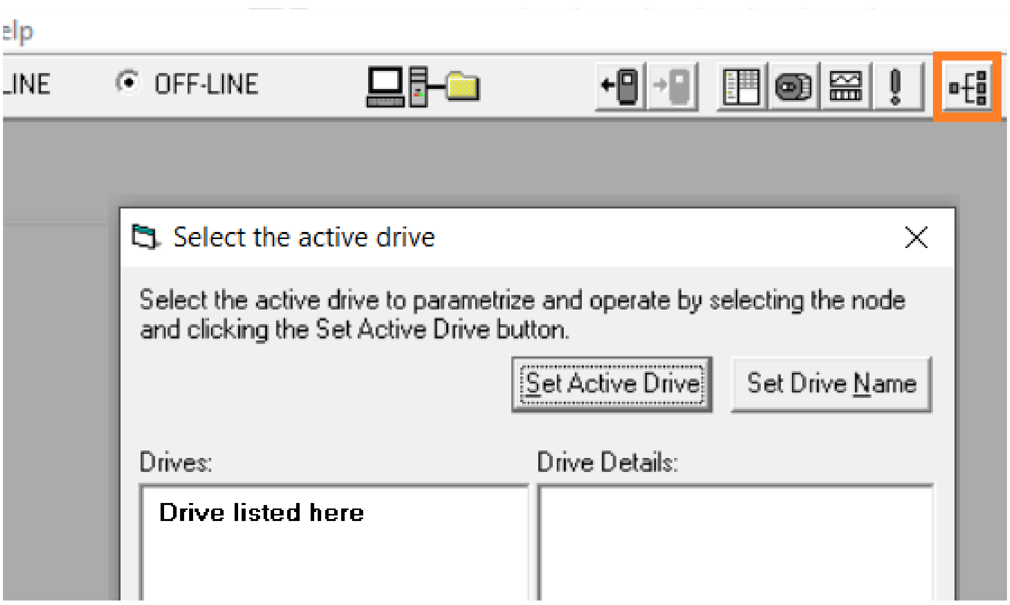
2.1. Remote installation via PLC:
- 2.1.1. Open remote desktop connection to PLC (10.4.0.40).
- 2.1.2. Install NCDrive program from the commissioning USB stick or Vacon website on PLC windows installation
- 2.1.3. After installing NCDrive, copy the VCN files in the configfiles folder of the V1.5.0 repository to “C:/NCEngine/Applications” on the PLC.
- 2.1.4. Start the NCDrive program on the PLC
- 2.1.5. Go to step 3

2.2. Local installation with wired connection
- 2.2.1. Connect the laptop to the OT network of the ePU (192 side)
- 2.2.2. Start the NCDrive program on the laptop

2.3. Connect to the Vacon drive
- 2.3.1. In NCDrive, go to Tools → options → Communication and select “ethernet” and “scan drives from Ethernet and use drive select window to select the active drive”.



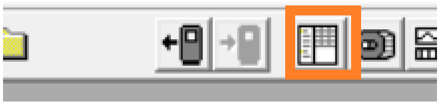
- 2.3.2. Press Apply/Ok
- 2.3.3. Click to the drive select button in the upper bar
- 2.3.4. The drive should be listed in the popup window, set as active drive and close the window



2.3.5. Toggle the system from “Offline” to “online”

2.4. Get the Inverter license keys

2.4.1. Open the parameter list in the top bar



2.4.2. In the parameter window scroll down to parameter p2.17.1 (gridcode license) and p3.2 (ugrid license) and write down both values. These will need to be entered again later.

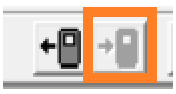
2.5. Open the new Parameter set

2.5.1. Go to file → open and open the Vacon_GENGRID_V1.2.Par file from the V1.5.0 update repository

2.5.2. Again, scroll down to parameter p2.17.1 (gridcode license) and p3.2 (ugrid license) and overwrite the current value with the values from step 4.2.

2.5.3. Also, scroll down to parameter p2.6.1.1 (Current limit) and check if this is set to 650A. If not, change this setting.

2.5.4. Click the “download” button to push the new settings into the Drive



If a message occurs stating “7 parameters could not be downloaded” you can ignore this.

2.6. Close NCdrive and disconnect from the PLC (if applicable).

3. PLC update

The Twincat update part is largely similar to previous updates (connect to plc and activate configuration).Two extra things may be required though:

- Rectify PLC hardware configuration, to be performed before new software upload
- Check and correct Machine settings, to be performed after new software upload

3.1. Rectify PLC hardware configuration

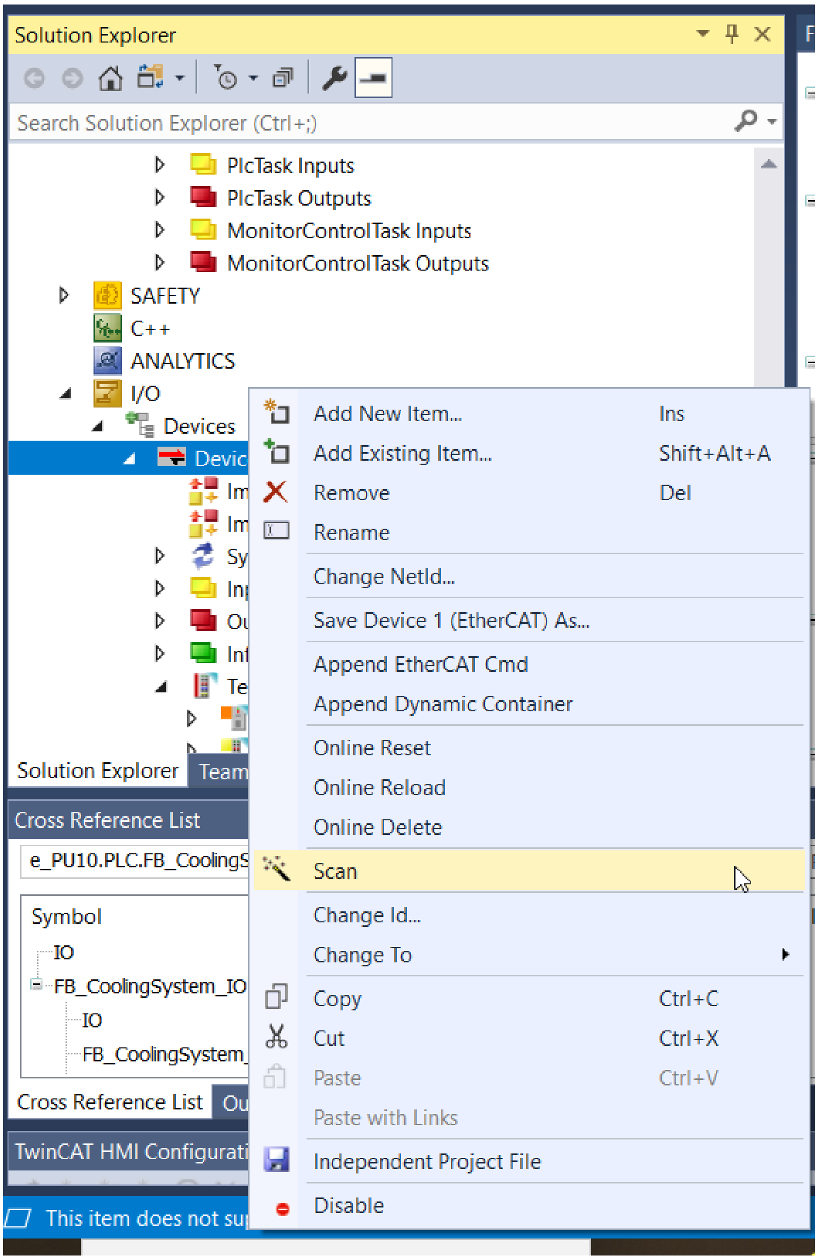
3.1.1. Open V1.5.0 twincat project and connect to the PLC

3.1.2. Select the right system variant (PN/MB, neutral, PSB, etc).

3.1.3. Check if the applicable Safety programs are activated/deactivated.

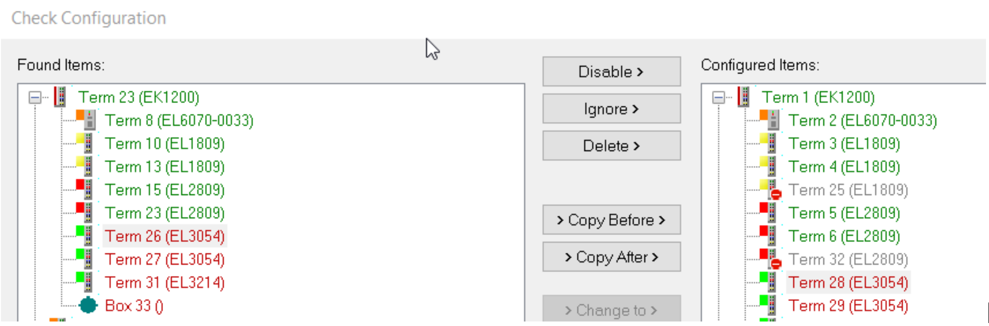
3.1.4. Put the PLC in config mode

3.1.5. Right click IO/Devices/Device 1 (EtherCAT) and select Scan



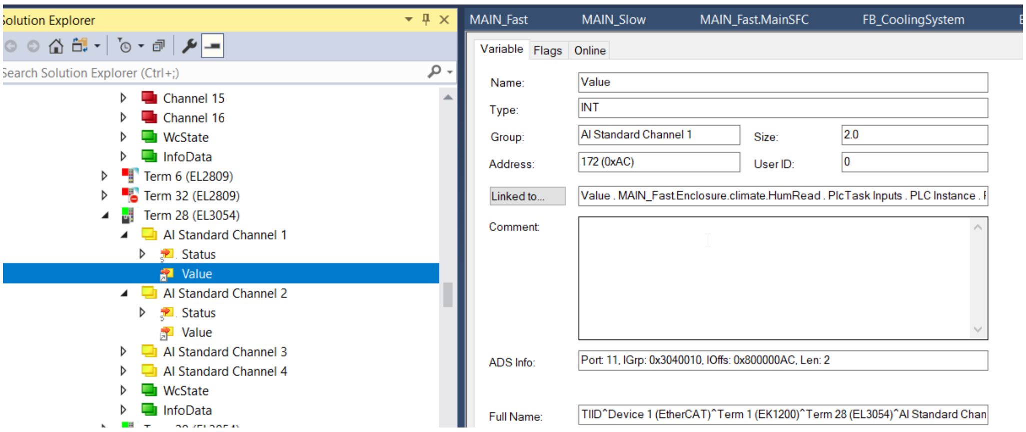
3.1.6. If a “the configuration is identical” message pops up, go to step 2

The EL3054 sometimes give errors on this (shown in red).

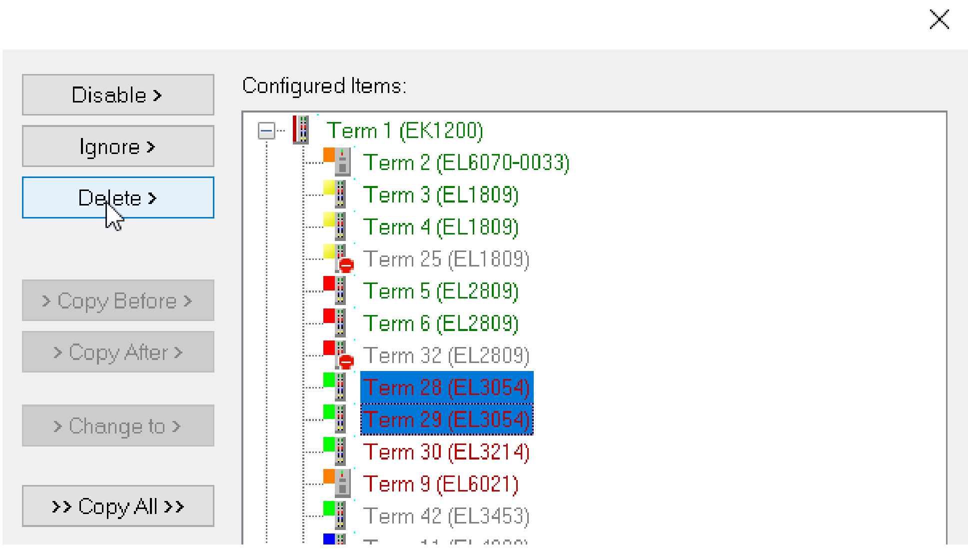


If this is the case:

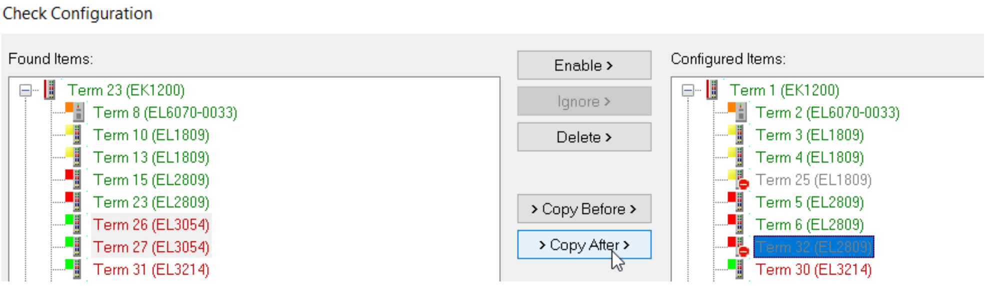
3.1.7. Close the scan window. Go to the EL3054 cards in the IO list on the left of the twincat main screen and write down which IO’s are connected to the EL3054 channels.



3.1.8. Then reopen the Scan window, delete the EL3054 cards from the right window.

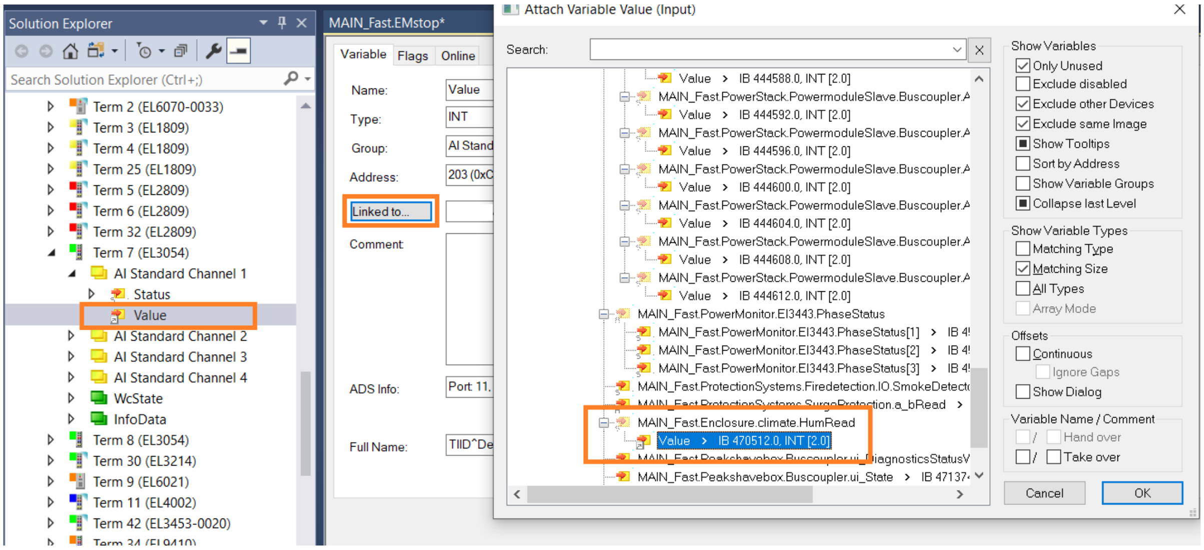


3.1.9. Then select the EL3054 cards in the left list and the last EL2809 card in the right and press “copy after”.



3.1.10. All cards should now be green or grey indicating the configuration is identical.

3.1.11. Go back to the EL3054 cards in the IO list in Twincat and reconnect the EL3054 inputs to the variables noted in 1.5.2:



3.2. Click “activate configuration” to push the new software to the PLC

3.3. Check and correct relevant hardware settings

3.3.1. “login” to the updated machine and go to:

e-PU10 → PLC → PLC → PLC Project → GVLs → Settings → Configuration

Check if all settings are correct for this machine. For example: AC pump feedback.